

Communication Protocols for the Internet of Things: A Few Choices

With more and more devices being connected to the Internet and millions of devices interacting with each other and the server, the need for communication protocols is critical. MQTT, CoAP, and Bluetooth are some of the communication protocols covered in this article.



In 2008, the number of connected devices in operation exceeded the number of humans connected to the Internet. It is estimated that by 2025 there will be more than 50 billion connected devices generating a revenue of US\$ 11 trillion. Though the term, the Internet of Things or IoT, was first coined back in 1999, the buzzword has started becoming a feasible reality in recent years. As we can see, the consumer electronics market is already flooded with smart and connected LED bulbs, home automation solutions and intelligent vehicles. Meanwhile, the Do-It-Yourself (DIY) hobbyist sector is seeing ultra-low power and high performance SoCs with built-in Wi-Fi, LoRa or Bluetooth communication features.

The prices of radio chips are now as low as US\$ 5 and there are tons of new products, unimaginable before but now a reality, as was seen at this year's Consumer Electronics Show (CES), Las Vegas and Mobile World Congress (MWC), Barcelona. Products like a smart toothbrush that learns your brushing habits, connected drones that can follow and record you while you are in the middle of an adventurous moment like river rafting, or a simple over-the-air (OTA) software update that can turn your car into a smart self-driving vehicle. With IoT and artificial intelligence backing it up, the possibilities are endless.

Getting started with IoT

IoT provides a great development stack, so everyone can contribute to its development and growth. It can be broadly

divided into three constituents.

1. *The hardware:* This makes up the 'things' part of IoT and usually has a small microcontroller with sensors/actuators and firmware running on it, which is responsible for how it functions. A good example of this would be a smart fitness tracker with, say, an ARM Cortex M4 microcontroller and an Inertial Measurement Unit (accelerometers or gyroscopes) sending data to your smartphone via Bluetooth.
2. *The software:* Firmware running on the device, mobile applications, cloud applications, databases, device management/implementation, the frontend to display data or an algorithm which gives intelligence to your IoT project—all come under the software portion of the IoT stack.



Figure 1: IoT - billions of devices connected with each other